Section 4.1 – 4.2 Exercises

Directions: For exercises 1 - 9, simplify the given expression. Write answers with positive exponents.

1. $\frac{a^3a^2}{a}$

2. $\frac{mn^2}{m^{-2}}$

3. $(b^3c^4)^2$

4. $\left(\frac{x^{-3}}{y^2}\right)^{-5}$

5. $(w^0x^5)^{-1}$

6. $\frac{m^4}{n^0}$

7. $y^{-4}(y^2)^2$

8. $\frac{p^{-4}q^2}{p^2q^{-3}}$

9. $(9z^3)^{-2}y$

Directions: For exercises 10 - 13, determine whether the equation represents exponential growth, exponential decay, or neither.

10. $y = 300(1 - t)^5$

11. $y = 220(1.06)^x$

12. $y = 16.5(1.025)^{\frac{1}{x}}$

13. $y = 11{,}701(0.97)^t$

14. Suppose an investment account is opened with an initial deposit of $\$12{,}000$ earning 7.2% interest compounded continuously. How much will the account be worth after 30 years?

15. How much less would the account from Exercise 14 be worth after 30 years if it were compounded monthly instead?

Directions: For exercises 16 - 22, evaluate each function. Round answers to four decimal places, if necessary.

16. $f(x) = 2(5)^x$, for $f(-3)$

17. $f(x) = -4^{2x+3}$, for $f(-1)$

18. $f(x) = e^x$, for $f(3)$

19. $f(x) = -2e^{x-1}$, for $f(-1)$

20. $f(x) = 2.7(4)^{-x+1} + 1.5$, for $f(-2)$

21. $f(x) = 1.2e^{2x} - 0.3$, for $f(3)$

22. $f(x) = -\frac{3}{2}(3)^{-x} + \frac{3}{2}$, for $f(2)$

23. An individual wants to save $54,000 for a down payment on a home. How much is needed to invest in an account with 8.2% APR, compounding daily, to reach $54,000 in 5 years?

24. An individual opened a retirement account with 7.25% APR in the year 2000. The initial deposit was $13,500. How much will the account be worth in 2025 if interest compounds monthly? How much more would they make if interest was compounded continuously?

25. An investment account with an annual interest rate of 7% was opened with an initial deposit of $4,000 Compare the values of the account after 9 years when the interest is compounded annually, quarterly, monthly, and continuously.

Directions: For exercises 26 - 28, graph the transformation of $f(x) = 2^x$. Give the horizontal asymptote, the domain, and the range.

26. $f(x) = 2^{-x}$

27. $h(x) = 2^x + 3$

28. $f(x) = 2^{x-2}$

Directions: For exercises 29 - 34, start with the graph of $f(x) = 4^x$. Then write a function that results from the given transformation.

29. Shift $f(x)$ 4 units upward

30. Shift $f(x)$ 3 units downward

31. Shift $f(x)$ 2 units left

32. Shift $f(x)$ 5 units right

33. Reflect $f(x)$ about the x-axis

34. Reflect $f(x)$ about the y-axis